



Data Article

FruitVision: A benchmark dataset for fresh, rotten, and formalin-mixed fruit detection



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ABSTRACT

This paper presents a large-scale, meticulously validated dataset designed for the classification of fruit quality across three critical categories: fresh, rotten, and formalin-mixed. This dataset provides a comprehensive collection of five fruit types—apple, banana, mango, orange, and grapes—classified into three categories. Ensuring fruit safety is essential, especially in detecting formalin-mixed samples. Formalin is a common preservative used to extend shelf life, but it poses serious health risks. The dataset was collected from April 18, 2024, to July 31, 2024, using three different mobile cameras to introduce diversity in image quality and conditions. Initially, the dataset comprised 10,154 images. To enhance its robustness and applicability, eight data augmentation techniques—rotation (i.e., 45, 60, 90 degree), horizontal flipping, zooming, brightness adjustment, shearing, and Gaussian noise addition—were applied, increasing the total number of images to 81,232. All images were uniformly resized to 512×512 pixels to ensure consistency and facilitate integration with machine learning and computer vision models. This dataset supports a variety of applications, including automated fruit classification, food quality control, and real-time monitoring through IoT technology. It offers valuable resources for research and development in automated detection systems, improving food safety and processing efficiency.

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and addressing the health risks associated with formalin exposure.

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Specifications Table

Subject	Computer Science
Specific subject area	Image Classification, Image Identification, Machine Learning, Deep Learning, and Computer Vision
Type of data	Image (.JPG)
Data collection	Image data was collected from local fruit markets using a range of mobile phone cameras. Fruits were categorized into three classes: fresh, rotten, and formalin-mixed. To create formalin-mixed images, fruits were soaked for 30 minutes in a solution of 10% formalin and 90% water. After soaking, the fruits were dried with a cloth and photographed two days later. This process ensured a diverse and comprehensive dataset for each fruit type, facilitating accurate classification across the three categories.
Data source location	Kodomtoli Fruit Market, Sylhet, Bangladesh Latitude: 24.8822° N, Longitude: 91.8738° E Sylhet Agricultural University (SAU), Sylhet, Bangladesh Latitude: 24.9101° N, Longitude: 91.9006° E Zone: Sylhet, Country: Bangladesh
Data accessibility	Repository name: Mendeley Data Data identification number: 10.17632/xkbjx8959c.2 Direct URL to data: https://data.mendeley.com/datasets/xkbjx8959c/2
Related research article	<i>None</i>

1. Value of the Data

- This dataset offers a unique and comprehensive collection of images classified into three categories: fresh, rotten, and formalin-mixed fruits. Formalin, a hazardous preservative used to extend the shelf life of fruits, poses significant health risks, making its detection crucial for public safety. The dataset provides a valuable resource for developing machine learning and deep learning models aimed at improving food safety, particularly in regions where chemical preservation is a concern.
- Containing 10,154 original images, with augmentation increasing the total to 81,232, this dataset has been enriched using eight augmentation techniques, including rotation, flipping, scaling, and brightness adjustment. This diversity ensures robustness, making it highly suitable for training advanced computer vision models. Additionally, this dataset includes a larger variety of fruit classes compared to existing datasets [1–3], enabling researchers to detect more fruit types and achieve better model performance.
- The dataset addresses the critical issue of formalin contamination, offering a valuable foundation for researchers in public health, toxicology, and food regulation. By providing a diverse set of formalin-mixed fruit images, it supports the development of models capable of detecting chemical preservatives, reducing the risks associated with formalin exposure.
- Beyond fruit classification, the dataset's structure and variety make it adaptable for a wide range of applications, including the development of Internet of Things (IoT) systems for real-time agricultural monitoring. Researchers can use the data to build smart sensors that detect spoilage or chemical contamination, contributing to advancements in smart farming technologies and enhancing food traceability across supply chains.

- This dataset can also be applied to broader research areas, such as computer vision, chemical analysis, and health informatics. Its versatility makes it a valuable resource for training and benchmarking algorithms across diverse fields, including object recognition, spoilage detection, and contamination analysis.
- The dataset includes 81,232 raw images of five common Bangladeshi fruits—apple, banana, orange, grape, and mango—captured from various angles to ensure diversity. Its unique inclusion of formalin-mixed samples supports research in food safety, enabling the development of AI models for detecting chemical contamination. It can be applied in use cases like automated quality control, smart agriculture, public health monitoring, and mobile-based food inspection tools.

2. Background

The primary motivation behind compiling this dataset was to address the critical issue of food safety, specifically in the context of detecting chemically mixed fruits, such as those preserved with formalin. Formalin is widely used in developing countries like Bangladesh to extend the shelf life of fruits, despite its harmful effects on human health. This dataset was developed to support research on automated detection of formalin-mixed fruits using machine learning and computer vision models. The intention was to create a diverse and comprehensive dataset that could aid in the identification and classification of fresh, rotten, and formalin-mixed fruits to improve food quality control. The dataset was generated through the collection of fruit images from local markets and gardens, utilizing multiple mobile cameras to capture diversity in angles, lighting, and quality. The methodological foundation includes the application of data augmentation techniques to enhance the dataset's usability for machine learning models. This data collection effort aligns with ongoing research on developing automated fruit quality assessment systems, and the dataset can serve as a valuable resource for improving food safety and monitoring practices in agricultural supply chains.

3. Data Description

The fruit market plays a pivotal role in both public health and the economy of Bangladesh. Fresh fruits, known for their nutritional value and flavor, are vital to maintaining population health and supporting local farmers and markets. However, the sale of rotten or chemically mixed fruits poses significant risks, contributing to long-term illnesses and food poisoning, which strain the healthcare system. The widespread use of chemicals like formalin to artificially preserve fruits also undermines consumer trust, limits export opportunities, and weakens the agricultural sector's contribution to the national economy. Ensuring the availability of uncontaminated, fresh fruits is crucial for public health and the economic stability of Bangladesh. This dataset provides a comprehensive collection of images of five fruit types—apple, banana, mango, orange, and grapes—classified into three categories: fresh, rotten, and formalin-mixed. The images were collected from various fruit markets and gardens across Sylhet, Bangladesh, between April 18, 2024, and July 31, 2024. A total of 10,154 images were captured using three different mobile cameras to ensure diversity in angles, lighting, and quality. To enhance the dataset, eight augmentation techniques, such as rotation, flipping, brightness adjustment, and noise addition, were applied, resulting in a final set of 81,232 images resized to 512×512 pixels.

Table 1 details the number of images collected and augmented for each fruit category, while Table 2 provides a description of each fruit type in the dataset. With over 80,000 images across the three classes, this dataset offers significant potential for improving both economic stability and public health through the application of machine learning and computer vision models. These models can be trained to develop automated systems that can quickly and accurately assess the quality of fruits, ensuring that only fresh and uncontaminated fruits reach consumers throughout the supply chain.

Table 1
Category-wise data distribution with augmentation technique.

Category	Number of original images	Number of augmented images
Fresh Apple	765	6120
Rotten Apple	630	5040
Formalin-mixed Apple	643	5144
Fresh Banana	749	5992
Rotten Banana	632	5056
Formalin-mixed Banana	660	5280
Fresh Grape	770	6160
Rotten Grape	630	5040
Formalin-mixed Grape	610	4880
Fresh Mango	763	6104
Rotten Mango	630	5040
Formalin-mixed Mango	616	4928
Fresh Orange	753	6024
Rotten Orange	656	5248
Formalin-mixed Orange	647	5176
Total	10,154	81,232

Table 3 provides a summary of the key metadata attributes associated with the fruit image dataset, including fruit class, capturing device, image resolution range, weather conditions, temperature, and the location where each image was collected.

Table 4 presents a comparative analysis between our proposed dataset and several existing fruit image datasets. While previous works by Sultana et al. [4], Abedin et al. [5], and Saha et al. [6] focus primarily on fresh and rotten fruit classes, none include formalin-mixed samples—a critical aspect in food safety. Moreover, our dataset offers a larger number of original images (10,154) and a significantly higher volume of total images after augmentation (81,232), compared to 3200–5470 original and up to 28,500 augmented images in prior datasets. In terms of class variety, our dataset covers 15 distinct categories across five common fruit types with three classification labels: fresh, rotten, and formalin-mixed—making it unique and more comprehensive for real-world applications in food quality detection and chemical contamination analysis.

4. Experimental Design, Materials and Methods

The research design for the fruit dataset followed a structured approach, beginning with the capture of fruit images using three mobile cameras to introduce diversity in lighting, angles, and image quality. The images were then classified into three distinct categories: fresh, rotten, and formalin-mixed. Agricultural experts validated the dataset to ensure the accuracy of classifications. Eight data augmentation techniques were applied to enhance the dataset, followed by resizing the images to 512 × 512 pixels for consistency and ease of use in machine learning models. The process of dataset preparation and classification, leading to integration into an expert system, is illustrated in Fig. 1.

4.1. Camera Specification

To capture the fruit images, three different mobile phones—Apple iPhone 15 Pro Max, Redmi POCO M2, and Redmi Note 9 Pro—were used, each equipped with high-resolution cameras. The varying camera specifications provided different image dimensions and introduced a diversity of perspectives, lighting, and image quality. This variation reflects real-world scenarios, enhancing

Table 2
Class-wise data description with visualization.

Class Name	Description	Visualization
Fresh Apple	A fresh apple has a tangy yet sweet flavour and is crisp and juicy. It is an antioxidant- and vitamin-rich food because it is high in vitamin C. When they are firm to the touch, flawless, and have a crisp bite to each mouthful, fresh apples are at their best.	
Rotten Apple	An apple that has gone bad will usually be squishy, discoloured, and smell bad. It can have black, mushy patches or mold, which are indicators that it's starting to deteriorate. Because of the bacteria and toxins, it contains, eating a rotten apple can be dangerous.	
Formalin-mixed Apple	Apples mixed with formalin, a solution containing formaldehyde, to artificially lengthen their shelf life are known as formalin-mixed apples. Appearing fresh and sturdy, these apples are harmful to eat because they contain formalin, a toxic substance that can cause cancer and other significant health problems.	
Fresh Banana	A fresh banana is a yellow, ripe fruit with a smooth texture and sweet flavour. Bananas are not only tasty but also good for your heart, stomach, and energy levels because they are full of vital nutrients like potassium, vitamin C, and dietary fibre. They can be added to other meals or eaten on their own as a quick and healthy snack.	
Rotten Banana	A banana that has become overripe and degraded is said to be rotten. It usually has mushy skin, a strong, sweet smell, and dark brown or black skin. The flesh of the banana softens and could have dark specks or streaks. Although rotted bananas are frequently too ripe to eat raw.	
Formalin-mixed Banana	A banana that has been mixed with formalin, a chemical solution containing formaldehyde, to maintain its look and increase its shelf life is known as a formalin-mixed banana. Despite their apparent freshness, eating such bananas is dangerous because they contain formalin, which can have detrimental effects on the respiratory system and possibly cause long-term health problems.	
Fresh Grape	Green, crimson, and purple are just a few of the hues that characterize fresh grapes, which are small, spherical fruits. They are full of vitamins, antioxidants, and water, and they taste sweet and juicy. Grapes can be used in baking and cooking, or they can be consumed on their own or mixed into salads. They serve as the foundation for goods like raisins and wine.	
Rotten Grape	When a grape deteriorates because of bacterial or fungal growth, it is said to be rotten. It usually turns discoloured, soft, and shrivelled; it also frequently develops mould or an unpleasant odour. If eaten, rotten grapes can lead to foodborne illnesses and should not be consumed. To keep grapes from becoming bad, they must be stored properly and consumed quickly.	
Formalin-mixed Grape	A grape that has been soaked in a formalin solution to maintain its appearance and lengthen its shelf life is known as a formalin-mixed grape. These grapes may seem fresh but eating them might be dangerous since formalin has poisonous properties that can lead to long-term health hazards and intestinal problems.	

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Table 2 (continued)

Class Name	Description	Visualization
Fresh Mango	Fresh mangoes are juicy, colourful fruits with a deep, fragrant aroma and a sweet, tropical flavour. When ripe, their silky, golden skin gives way to the juicy, vitamin-rich orange flesh within. Mangoes are delicious and provide many health benefits, including helping digestion and strengthening the immune system. They are often eaten raw or added to smoothies, salads, and desserts.	
Rotten Mango	A rotten mango has a mushy, soft feel and a bad, sour fragrance. The fruit might have a discernible mold development, and the skin could be discoloured or blackened. Overripening, inappropriate storage, or pathogen exposure are the reasons for the deterioration. Mangoes that have gone bad should be thrown away since they may contain dangerous bacteria and be unsafe to consume.	
Formalin-mixed Mango	Mangos are mixed with formalin, a chemical that extends their freshness and shelf life. These mangoes may seem ripe and delicious, but they have serious health hazards because formalin is poisonous and can lead to serious conditions like cancer and digestive disorders.	
Fresh Orange	Juicy and vivid, fresh oranges are a great source of antioxidants and vitamin C. They are a versatile element in both sweet and savoury meals and a refreshing snack due to their tangy-sweet flavour. Fresh oranges are nutrient-dense and promote general health by boosting the immune system, enhancing skin health, and facilitating digestion.	
Rotten Orange	Dark patches, a sour scent, and a squishy, mushy feel are classic indicators that an orange is rotten. The fruit within may taste bad or be discoloured, and the skin may seem shrivelled or discoloured. Rotting is caused by the growth of bacteria or mold, which is frequently hastened by unsuitable storage conditions. To minimize health concerns and stop mold from spreading to other fruits, it is better to throw away rotten oranges.	
Formalin-mixed Orange	To keep an orange's appearance and prolong its shelf life, it is mixed with formalin, a chemical solution that contains formaldehyde. Despite their seemingly fresh appearance, eating such oranges is dangerous since formalin can have harmful long-term consequences as well as toxicity.	

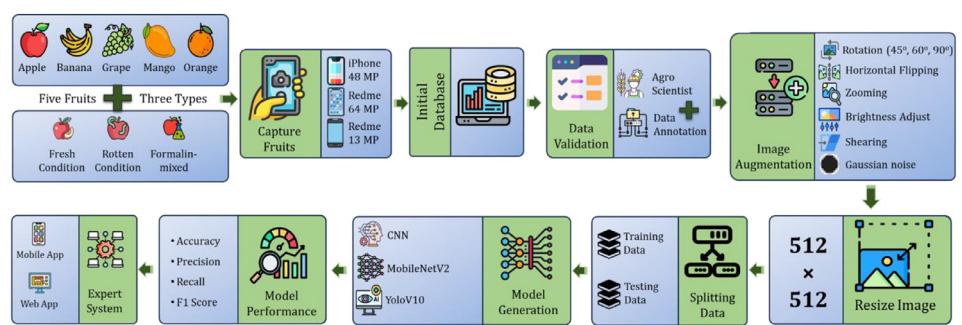


Fig. 1. Step by step working procedure of fruits and its condition classification.

Table 3
Dataset attributes include class, device, resolution, weather, temperature, and location (Note: WB = White Background).

Class	Device	Image Resolution (Range)	Weather	Temperature	Location
Fresh Apple	Apple iPhone 15	840 to	Sunny	32° C	Sylhet (WB)
	Pro Max (60%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (10%)	780 to 1280 × 960			
	Redmi Note 9 Pro	to 2160			
Rotten Apple	Apple iPhone 15	840 to	Sunny	30° C	Sylhet (WB)
	Pro Max (50%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (20%)	780 to 1280 × 960			
	Redmi Note 9 Pro	to 2160			
Formalin-mixed Apple	Apple iPhone 15	840 to	Sunny	30° C	Sylhet (Laboratory)
	Pro Max (60%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (30%)	780 to 1280 × 960			
	Redmi Note 9 Pro	to 2160			
Fresh Banana	Apple iPhone 15	840 to	Foggy	24° C	Sylhet (WB)
	Pro Max (50%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (30%)	780 to 1280 × 960			
	Redmi Note 9 Pro	to 2160			
Rotten Banana	Apple iPhone 15	840 to	Sunny	34° C	Sylhet (WB)
	Pro Max (60%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (10%)	780 to 1280 × 960			
	Redmi Note 9 Pro	to 2160			
Formalin-mixed Banana	Apple iPhone 15	840 to	Sunny	32° C	Sylhet (Laboratory)
	Pro Max (55%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (25%)	780 to 1280 × 960			
	Redmi Note 9 Pro	to 2160			
Fresh Grape	Apple iPhone 15	840 to	Foggy	26° C	Sylhet (WB)
	Pro Max (45%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (30%)	780 to 1280 × 960			
	Redmi Note 9 Pro	to 2160			
Rotten Grape	Apple iPhone 15	840 to	Sunny	31° C	Sylhet (WB)
	Pro Max (45%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (20%)	780 to 1280 × 960			
	Redmi Note 9 Pro	to 2160			
Class	Device	Image Resolution (Range)	Weather	Temperature	Location
Rotten Grape	Apple iPhone 15	840 to	Sunny	31° C	Sylhet (WB)
	Pro Max (45%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (20%)	780 to 1280 × 960			
	Redmi Note 9 Pro	to 2160			
		840 to 1080 × 1260			
		to 3280			

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Table 3 (continued)

Class	Device	Image Resolution (Range)	Weather	Temperature	Location
Formalin-mixed Grape	Apple iPhone 15	840 to	Sunny	30° C	Sylhet (Laboratory)
	Pro Max (50%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (10%)	780 to 1280 × 960			
	Redmi Note 9 Pro (40%)	to 2160			
Fresh Mango	Apple iPhone 15	840 to 1080 × 1260	Sunny	31° C	Sylhet (WB)
	Pro Max (40%)	to 3280			
	Redmi POCO M2	3000 × 1260 to			
	Reloaded (30%)	3000			
	Redmi Note 9 Pro (30%)	780 to 1280 × 960			
Rotten Mango	Apple iPhone 15	to 2160	Sunny	32° C	Sylhet (WB)
	Pro Max (70%)	840 to 1080 × 1260			
	Redmi POCO M2	to 3280			
	Reloaded (10%)	3000 × 1260 to			
	Redmi Note 9 Pro (20%)	3000			
Formalin-mixed Mango	Apple iPhone 15	780 to 1280 × 960	Foggy	27° C	Sylhet (Laboratory)
	Pro Max (40%)	to 2160			
	Redmi POCO M2	840 to 1080 × 1260			
	Reloaded (25%)	to 3280			
	Redmi Note 9 Pro (35%)	3000 × 1260 to			
Fresh Orange	Apple iPhone 15	3000	Sunny	34° C	Sylhet (WB)
	Pro Max (65%)	780 to 1280 × 960			
	Redmi POCO M2	to 2160			
	Reloaded (15%)	840 to 1080 × 1260			
	Redmi Note 9 Pro (20%)	to 3280			
Rotten Orange	Apple iPhone 15	840 to	Sunny	33° C	Sylhet (WB)
	Pro Max (60%)	3000 × 1260 to			
	Redmi POCO M2	3000			
	Reloaded (20%)	780 to 1280 × 960			
	Redmi Note 9 Pro (20%)	to 2160			
Formalin-mixed Orange	Apple iPhone 15	840 to 1080 × 1260	Sunny	30° C	Sylhet (Laboratory)
	Pro Max (60%)	to 3280			
	Redmi POCO M2	3000 × 1260 to			
	Reloaded (30%)	3000			
	Redmi Note 9 Pro (10%)	780 to 1280 × 960			
		to 2160			
		840 to 1080 × 1260			
		to 3280			

the dataset's applicability for image processing tasks. Detailed camera specifications for these devices are presented in [Table 5](#).

4.2. Geographical Location

The fruit image dataset was systematically collected through extensive field surveys conducted in Bangladesh between April 18, 2024, and July 31, 2024. The selection of collection

Table 4
Comparison with available dataset on fruits.

Dataset	Class	Number of Fruits	Num. of original images	Num. of augmented images	Fruits Type (Class)		
					Fresh	Rotten	Formalin mix
Sultana et al. [4]	16	8	3,200	12,335	✓	✓	×
Abedin et al. [5]	10	10	3,173	×	✓	×	×
Saha et al.[6]	7	7	5,470	28,500	✓	×	×
This dataset	15	5	10,154	81,232	✓	✓	✓

Table 5
Details of camera specification.

Mobile Model Name	Camera Type	Resolution (MP)	Focal Length (mm)	Sensor Size (inch)
Apple iPhone 15 Pro Max	Main (Wide)	48	24	1/1.28
	Telephoto	12	120	1/3.06
	Ultrawide	12	13	1/2.55
Redmi POCO M2 Reloaded	Main (Wide)	13	28	1/3.10
	Ultrawide	8	13	1/4.0
	Macro	5	1.3	1/5.0
Redmi Note 9 Pro	Main (Wide)	64	26	1/1.72
	Ultrawide	8	13	1/4.0
	Macro	5	1.3	1/5.0

sites was strategically planned to encompass diverse geographical, environmental, and agricultural conditions, ensuring a well-balanced and representative dataset. The geographic locations of the fruit data collection sites are as follows:

1. Kodomtoli Fruit Market, Sylhet, Bangladesh
Latitude: 24.8822° N, **Longitude:** 91.8738° E
2. Sylhet Agricultural University (SAU), Sylhet, Bangladesh
Latitude: 24.9101° N, **Longitude:** 91.9006° E

4.3. Data Validation and Annotation

The data validation and annotation process involved careful inspection by agricultural experts to ensure the accuracy of fruit classifications into fresh, rotten, and formalin-mixed categories (The validation certification is given as supplementary file). After capturing the images, experts conducted visual inspections and chemical tests to verify the formalin treatment. Each image was annotated with corresponding labels, ensuring consistency across the dataset. The detailed steps of this validation and annotation process are outlined in Algorithm 1, which illustrates the systematic approach followed to guarantee the dataset’s reliability.

4.4. Preprocessing

To develop an expert system using machine learning and deep learning models with our dataset, this study followed a structured preprocessing workflow consisting of five key steps: (a) Data Augmentation, (b) Image Resizing, (c) Dataset Splitting, (d) Model Generation, (e) Performance Analysis, and (f) Expert System Integration, as visualized in Fig. 1. Given the need for a large volume of data to train deep learning models, this study employed augmentation tech-

Algorithm 1

Step by step data validation process with domain expert.

Input:	Dataset D containing N (i.e., 10,154) sample of 5 fruits in 3 conditions (fresh, rotten, formalin-mixed) and domain expert E
Output:	Validated and annotated dataset D'

Step-1:	Let, initial dataset D and set of classification condition $C = \{\text{fresh, rotten, formalin-mixed}\}$
Step-2:	Define classification criteria by domain expert E and set of criteria in a document T
Step-3:	Randomly sampling from a subset D_s from D where $D_s \subset D$ and $ D_s = m$ (E.g., 20% of N)
Step-4:	FOR each $d_i \in D_s$
Step-5:	Present d_i to the domain expert E
Step-6:	Let $E(d_i)$ be the label assigned by the expert
Step-7:	Store the expert's classification results in a list $L = \{E(d_i)\}_{i=1}^m$
Step-8:	ENDFOR
Step-9:	FOR each sample $d_i \in D_s$
Step-10:	Compare the expert label $E(d_i)$ with the dataset label $D(d_i)$
Step-11:	Record the comparison results in a list $R = \{r_i\}_{i=1}^m$
Step-18:	IF $E(d_i) = D(d_i)$
Step-19:	Then, set $r_i = 1$
Step-20:	ELSE
Step-21:	Then, set $r_i = 0$
Step-22:	ENDIF
Step-23:	ENDFOR
Step-24:	Analyze discrepancies and identify sample where $E(d_i) \neq D(d_i)$
Step-25:	Discuss discrepancies with the domain expert E to understand reasons for misclassifications and adjust the classification criteria T
Step-26:	Calculate accuracy, $A = \frac{\sum_{i=1}^m r_i}{m}$
Step-27:	FOR each $d_i \in D_s$
Step-28:	IF $E(d_i) \neq D(d_i)$
Step-29:	Then, update label D to $E(d_i)$
Step-30:	ENDIF
Step-31:	ENDFOR
Step-32:	Repeat steps 3 to 30 with new random subsets D_s until the validation metrics stabilize and expert satisfied with the dataset
Step-33:	Finalize the validate dataset D'

niques to enhance the dataset. Augmentation included random rotations (45°, 60°, and 90°), horizontal flip (0.5), scaling/zooming (0.8 to 1.2), shearing (-16, 16), brightness adjustment (0.8 to 1.2), and additive gaussian noise(scale=0.05 × 255). These transformations introduced variability to the dataset, making it more robust for model training. Some sample augmented images are presented in Fig. 2.

The collected images were initially diverse in size and shape, as they were captured from various fruit markets and gardens. To standardize the dataset, resized all images to 512 × 512 pixels using bicubic interpolation. The 'fill_mode()' function from the Keras package in Python was utilized for parameter estimation during resizing, ensuring image consistency across the dataset. This uniform size allows the images to be efficiently processed by deep learning architectures.

Once resized, the dataset was split into training and testing sets, preparing it for model development. Future work will explore different deep learning models to determine the most effective method for real-world applications, particularly in fruit quality assessment and chemical detection.

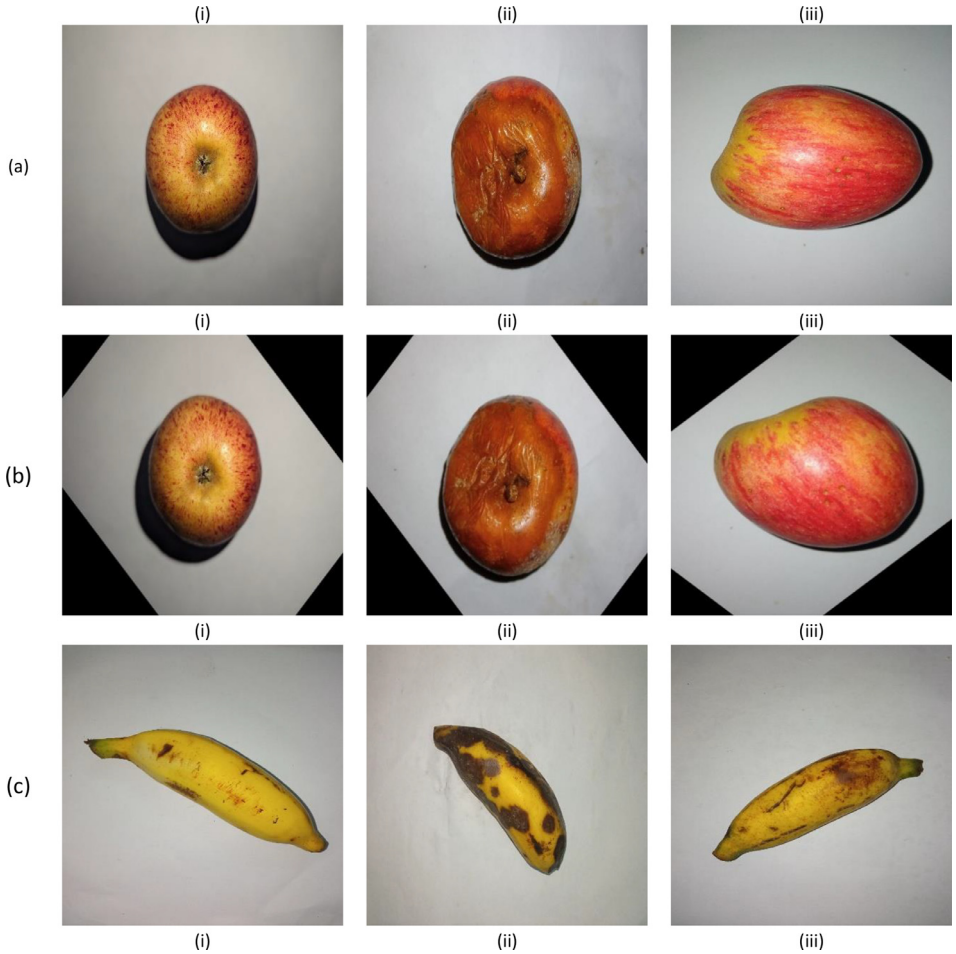


Fig. 2. Sample images of fruits and their augmented counterparts for three types: (i) fresh, (ii) rotten, and (iii) formalin-mixed. (a) Apple, (b) Augmented Apple, (c) Banana, (d) Augmented Banana, (e) Grape, (f) Augmented Grape, (g) Mango, (h) Augmented Mango, (i) Orange, (j) Augmented Orange.

4.5. Model Implementation

For classification of five fruit types into fresh, rotten, and formalin-mixed classes, this study employ CNN, MobileNetV2, and YOLOv10 models with different computational and application needs in mind. The dataset was split in an 80:10:10 ratio for training, validation, and testing respectively. CNN is employed as a baseline for spatial feature capture through convolutional layers and multi-class classification by a softmax classifier. MobileNetV2 uses transfer learning for higher efficiency through depthwise separable convolutions and inverted residual blocks, with optimization for lightweight applications. YOLOv10 for real-time detection uses an anchor-free detection approach and CSPDarkNet backbone to address accurate localization and classification. For CNN and MobileNetV2, we used a learning rate of 0.001 with the Adam optimizer, batch size of 32, and trained for 50 epochs using 512×512 input images. For YOLOv8 (YOLOv8n-cls), the model was trained with a learning rate of 0.01, batch size of 16, and 100 epochs using the SGD optimizer. All models were trained and tested on Google Colab Pro with an NVIDIA Tesla

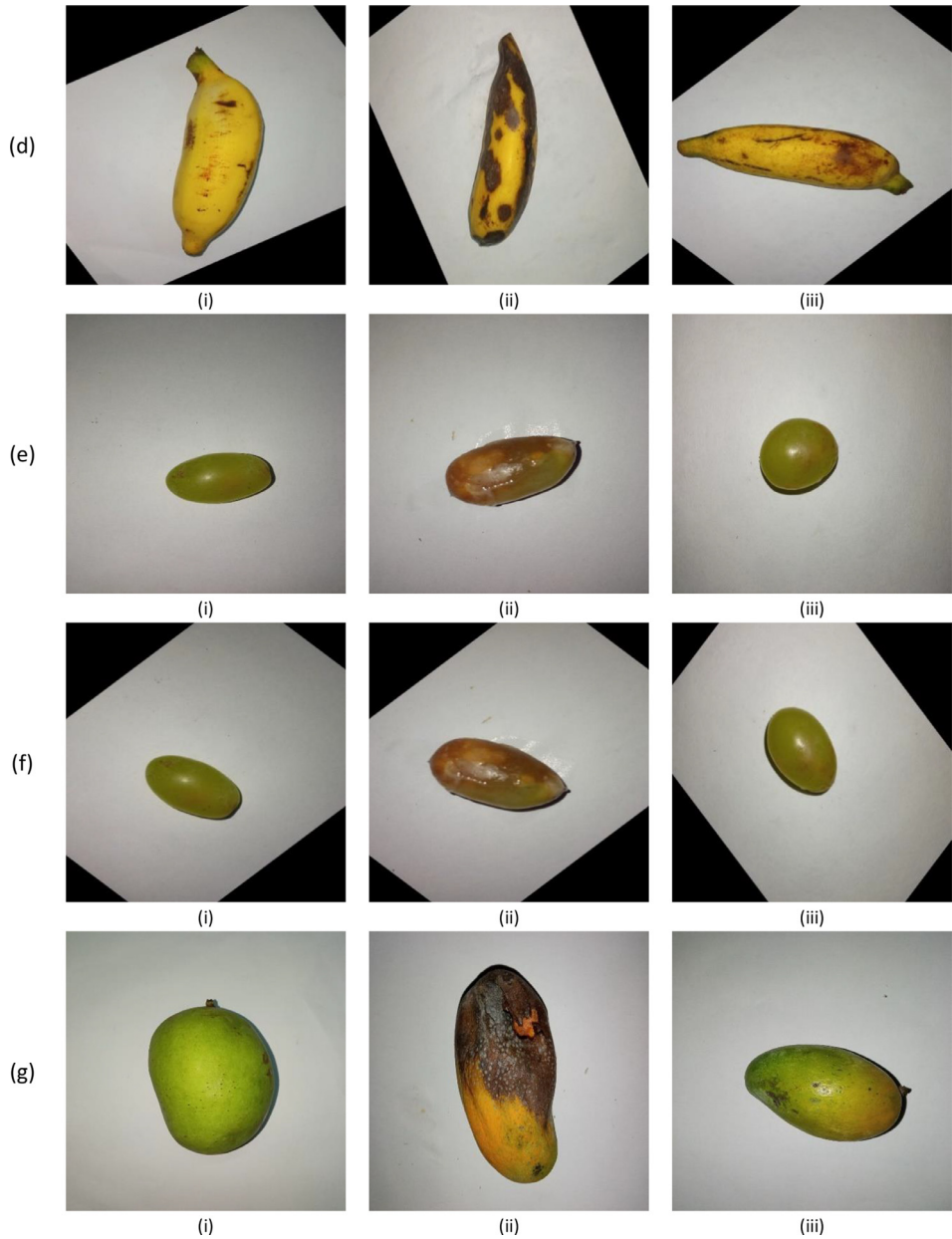


Fig. 2. Continued

T4 GPU (16 GB VRAM) and 32 GB RAM, using Python 3.10 with TensorFlow, PyTorch, OpenCV, and the Ultralytics YOLOv8 framework. All the

According to Table 6, among the three models, YOLOv10 is the most accurate at $94.46 \pm 2 \%$, being most appropriate for real-time fruit classification, and MobileNetV2 ($91.69 \pm 3\%$) provides a lightweight, computationally efficient alternative. CNN ($83.90 \pm 2 \%$) is reasonably good but lacks the advanced feature extraction capability of the other two. The final model selection will

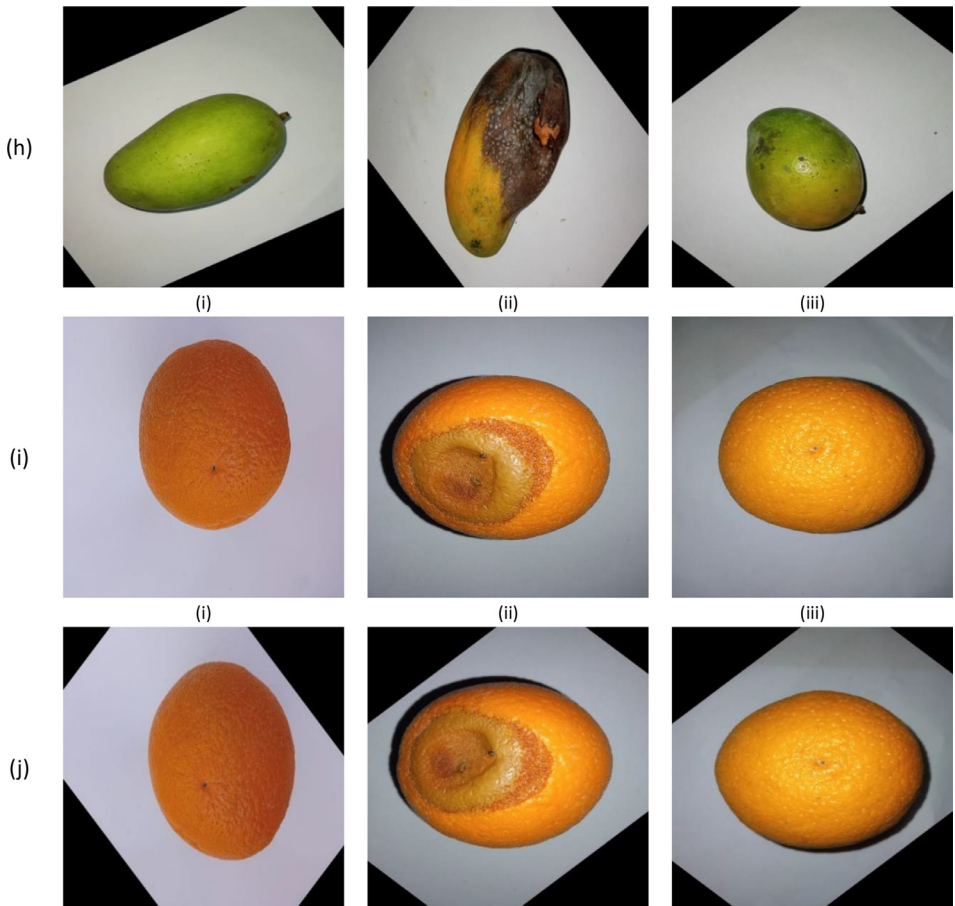


Fig. 2. Continued

depend on the accuracy-computational complexity-real time processing trade-off. The complete dataset, along with metadata, augmentation scripts, and usage instructions, is publicly available in our GitHub repository [7].

Regarding data augmentation, we applied commonly used techniques to increase dataset diversity and improve model generalization under varying conditions. While we recognize that certain augmentations may introduce slight distributional shifts, we carefully selected parameters to reflect real-world variability. Importantly, both original and augmented images are provided in separate folders, allowing researchers to choose or adjust augmentation based on their specific goals. This design ensures flexibility and supports robust experimentation while preserving data integrity.

In addition to its value in academic research, the dataset offers substantial potential for real-world applications, particularly in supply chain monitoring and food safety. In cold-chain logistics, it can be integrated with existing IoT frameworks—utilizing sensors and cameras alongside machine learning models—for real-time fruit classification. These systems can monitor environmental parameters such as temperature, humidity, and gas levels during transport and storage, enabling early detection of spoilage, contamination, or formalin adulteration. The dataset's inclusion of formalin-mixed samples further enhances its utility in regulatory inspections aimed at curbing chemical adulteration.

Table 6
Performance of applied deep learning models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	AUC
CNN	83.90 ± 2	81.12 ± 2	81.02 ± 2	81.07 ± 2	0.86 ± 0. 02
MobileNet-V2	91.69 ± 3	89.93 ± 3	89.28 ± 2	89.61 ± 2	0.91 ± 0.02
YOLO-V10	94.46 ± 2	93.62 ± 2	94.37 ± 3	93.99 ± 3	0.96 ± 0.03

Moreover, the dataset can be deployed in smart inspection systems to ensure quality assurance, reduce food waste, and support cloud-based platforms that notify stakeholders of any quality deviations. On the consumer side, the dataset facilitates mobile application development that empowers users to assess fruit quality using smartphone cameras—promoting informed purchasing decisions and enhancing public health awareness, especially in regions where food adulteration is prevalent.

Limitations

The dataset provides a substantial volume of images across five commonly consumed fruits (apple, banana, mango, orange, and grapes), one limitation is the restricted variety of fruit types, which may limit its applicability to studies involving a broader range of produce. Additionally, the use of mobile cameras with varying resolutions and capture conditions (e.g., lighting, background) may introduce potential biases affecting model consistency and robustness. The dataset's focus on tropical fruits prevalent in Bangladesh also presents challenges in generalizing findings to fruits from different regions or climates. Moreover, the absence of temporal data restricts the ability to analyze fruit spoilage progression over time, which could be valuable for time-series-based models in food quality monitoring.

Ethics Statement

All procedures for the fruit data collection were conducted in compliance with Daffodil International University's (DIU) ethical guidelines and relevant regulations. Ethical approval was granted by the Research Ethics Committee, Faculty of Science and Information Technology, DIU, under the approval number REC-FSIT-2024-04-17. This approval followed a comprehensive review process, ensuring adherence to safety protocols and ethical standards. The formalin-mixed fruits were strictly used for research purposes, validated by agricultural experts, and were not intended for consumption or sale to consumer.

CRedit Author Statement

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Data Availability

FruitVision: A Benchmark Dataset for Fresh, Rotten, and Formalin-mixed Fruit Detection (Original data) (Mendeley Data).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary Materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.dib.2025.111752](https://doi.org/10.1016/j.dib.2025.111752).

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